

Phase_2_Processing_Italy_SII_demo

October 5, 2025

1 Italian Solvency reports Table S.02.01.02; Phase 2 Processing

This script is a continuation of `Phase_1_Extraction_Italy_SII_demo`. It shows the second Phase of this process. The processing phase applies different transformations to the raw OCR data from the previous phase. Examples of transformations are changing separators, removing trailing zeros, remove dolar symbols etc.

1.1 Companies in scope

For the year 2024, the companies in scope are the following:

- Credemvita S.p.A.
- AXA MPS Assicurazioni Vita
- CRÈDIT AGRICOLE VITA
- Società Reale Mutua di Assicurazioni
- Cardif Vita S.p.A.
- MEDIOLANUM VITA S.p.A.
- Generali Italia S.p.A.
- Banco BPM Vita S.p.A.
- HDI ASSICURAZIONI S.p.A.
- Gruppo Assicurativo Poste Vita
- FIDEURAM VITA S.P.A.
- CNP Vita Assicura S.p.A.
- ITAS VITA
- Helvetia Vita S.p.A.
- Vittoria Assicurazioni S.p.A.
- GROUPAMA ASSICURAZIONI S.P.A.
- UniCredit Allianz Vita S.p.A.
- Zurich Investments Life S.p.A.

1.2 Description of the process

The process of extraction is performed in 5 phases:

1.2.1 Phase 0: Find the reports and identify the relevant tables.

- 1) Identify the new SFCR report and save it into the folder Input.
- 2) Identify the pages where the tables of interest are.
- 3) Compile the map of the company run in the `master_list.csv`.

1.2.2 Phase 1: Run the Extraction script.

The script performs the following steps (with slight modifications depending on the table format):
1) Save the page with the table into a separate folder Single_pdf. 2) Use either a Python package or specialized LLM to create a digital equivalent of the table. 3) Fix the systemic errors that prevent the table from being saved as DataFrame. 4) Save the DataFrame into the Output folder.

1.2.3 Phase 2: Run the Processing script (this script).

The script applies fixes to the DataFrame to make the numbers closer to the reported numbers. It joins all the tables into a single dataset and saves it into the Dirty_Combined folder.

1.2.4 Phase 3: Run the Cross-Validation script.

Applies a series of tests that check for the internal consistency between the numbers. Flags potential errors. After the individual fixes are applied, it saves the table into the Cleaner_Combined folder.

1.2.5 Phase 4: Final modifications to the table and a manual inspection.

1.3 Python packages

```
[147]: import pandas as pd
import numpy as np
```

1.4 Functions

```
[148]: def remove_trailing_zeros(data: pd.DataFrame, col: str, n: int) -> pd.DataFrame:
    """
    Removes exactly n trailing zeros from numbers in a column, if present.

    Args:
        data (pd.DataFrame): Input dataframe.
        col (str): Column name to process.
        n (int): Number of trailing zeros to remove.

    Returns:
        pd.DataFrame: DataFrame with modified column.
    """
    def clean_value(x):
        if pd.isna(x):
            return x
        try:
            # Convert to int first (in case it's float or string with decimals)
            x_int = int(float(x))
            str_x = str(x_int)
            zeros = "0" * n
            if str_x.endswith(zeros):
                return int(str_x[:-n]) # remove trailing zeros
            return x_int
```

```

        except ValueError:
            return x # return unchanged if not a valid number

data[col] = data[col].apply(clean_value)
return data

```

```

[149]: def adjust_decimals(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Adjust values in a dataframe column:
        - If a value has non-zero decimal digits, multiply it by 1000.
        - Otherwise, leave it unchanged.

        Parameters
        -----
        data : pd.DataFrame
            Input dataframe with numeric values.
        col : str
            Column name to check (default "C0010").

        Returns
        -----
        pd.DataFrame
            Updated dataframe with adjusted values.
        """
        data = data.copy()
        decimals = data[col] % 1
        data[col] = np.where(decimals != 0, data[col] * 1000, data[col])
        return data

```

```

[150]: def remove_space(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Remove every space in a column of strings in a DataFrame.
        """
        data[col] = (
            data[col]
            .str.replace(" ", "", regex=False) # remove thousand separators
        )
        return data

```

```

[151]: def remove_dot_sep(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Remove every dot in a column of strings in a DataFrame.
        """
        data[col] = (
            data[col]
            .str.replace(".", "", regex=False)
        )

```

```
return data
```

```
[152]: def as_float(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Convert a column of strings in a DataFrame into floating numbers.
        """
        data[col] = (
            data[col]
            .astype(float)
        )
        return data
```

```
[153]: def remove_comma_sep(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Remove every comma in a column of strings in a DataFrame.
        """
        data[col] = (
            data[col]
            .str.replace(",", "", regex=False)
        )
        return data
```

```
[154]: def remove_dolar_sep(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Remove every dolar sign in a column of strings in a DataFrame.
        """
        data[col] = (
            data[col]
            .str.replace("$", "", regex=False)
        )
        return data
```

```
[155]: def replace_coma_with_dot(data: pd.DataFrame, col: str = "C0010") -> pd.
        ↪DataFrame:
        """
        Replace every comma in a column of strings of a DataFrame with a dot.
        """
        data[col] = (
            data[col]
            .str.replace(",", ".", regex=False)
        )

        return data
```

```
[156]: def fix_dot_numbers(data: pd.DataFrame, col: str) -> pd.DataFrame:
        """
        Fix numbers in a column where dots are used as thousand separators
```

but keep values like '0.000' unchanged.

Args:

data (pd.DataFrame): Input dataframe.

col (str): Column name to fix.

Returns:

pd.DataFrame: DataFrame with the cleaned column.

"""

```
def clean_value(x):
    if pd.isna(x): # Handle NaN
        return x
    x_str = str(x)
    before, _, after = x_str.partition(".")
    if before == "0" and after == "000":
        return x_str # keep '0.000'
    return x_str.replace(".", "") # remove dots otherwise

data[col] = data[col].apply(clean_value)
return data
```

```
[157]: def replace_dash_with_zero(data: pd.DataFrame, col: str) -> pd.DataFrame:
        """
        Replace cells containing only '-' with 0 in the specified column.

        Parameters
        -----
        data : pd.DataFrame
            Input dataframe.
        col : str
            Column name where replacement should happen.

        Returns
        -----
        pd.DataFrame
            DataFrame with '-' replaced by 0 in the specified column.
        """
        data = data.copy()
        data[col] = data[col].apply(lambda x: 0 if str(x).strip() == "-" else x)
        data[col] = data[col].apply(lambda x: 0 if str(x).strip() == "-" else x)
        return data
```

```
[158]: def remove_bracket_instead_negative(data: pd.DataFrame, column: str) -> pd.
        ↪ DataFrame:
        """
        Converts numbers stored as strings with brackets into negative numbers.
```

Example: "(123)" -> "-123"

Parameters:

data (pd.DataFrame): Input DataFrame

column (str): Column name where conversion should be applied

Returns:

pd.DataFrame: Updated DataFrame with cleaned column

"""

data = data.copy()

def convert(value):

if isinstance(value, str) and value.startswith("(") and value.
endswith(")"):↵

return "-" + value[1:-1] # remove brackets, prepend "-"

return value

data[column] = data[column].apply(convert)

return data

[159]: def initial_read(unique_id: str, list:pd.DataFrame) -> pd.DataFrame:

"""

Find the location of the table and load it.

"""

path = list.loc[unique_id, "output_final_path"]

return pd.read_csv(path, index_col = 0).fillna(0)

[160]: def shorten_index(data: pd.DataFrame) -> pd.DataFrame:

"""

Trims the index values so they contain only the first 5 characters.

Example:

'R0510 2.337.991' -> 'R0510'

"""

data = data.copy()

data.index = data.index.astype(str).str[:5]

return data

[161]: def convert_index_to_R(data: pd.DataFrame) -> pd.DataFrame:

"""

*Converts the dataframe index by replacing the first digit with 'R'
and formatting the rest as string without decimals.*

Example:

80000.0 -> "R0000"

80500.0 -> "R0500"

```

"""
data = data.copy()
data.index = (
    data.index.astype(int).astype(str).str[1:].str.zfill(4).map(lambda x:
↪ "R" + x)
)
return data

```

2 Master list of companies

```
[162]: master_list = pd.read_csv("master_list.csv", header=0, index_col=0)
```

```
[163]: display(master_list)
```

id	company \
CREDEM_VITA_02_1	CREDEM
CREDEM_VITA_02_2	CREDEM
AXA_VITA_02_01	AXA
CREDAG_VITA_02_01	CREDIT_AGRICOLE
CREDAG_VITA_02_02	CREDIT_AGRICOLE
REALE_02_01	REALE_MUTUA
REALE_02_02	REALE_MUTUA
CARDIF_02_01	CARDIF
MEDIO_02_01	MEDIOLANUM
MEDIO_02_02	MEDIOLANUM
GEN_ITA_02_01	GENERALI ITALIA
GEN_ITA_02_02	GENERALI ITALIA
BMP_VITA_02_01	BMP VITA
BMP_VITA_02_02	BMP VITA
HDI_01	HDI
HDI_02	HDI
POSTE_VITA_01	POSTE_VITA
POSTE_VITA_02	POSTE_VITA
INTESA_VITA_01	INTESA_VITA
INTESA_VITA_02	INTESA_VITA
CNP_VITA_01	CNP_VITA
CNP_VITA_02	CNP_VITA
ITAS_VITA_01	ITAS_VITA
ITAS_VITA_02	ITAS_VITA
HELVETIA_VITA_01	HELVETIA_VITA
HELVETIA_VITA_02	HELVETIA_VITA
VITTORIA_01	VITTORIA
VITTORIA_02	VITTORIA
GROUPAMA_01	GROUPAMA
ALLIANZ_UNICREDIT_01	ALLIANZ_UNICREDIT
ALLIANZ_UNICREDIT_02	ALLIANZ_UNICREDIT

ZURICH_LIFE_01
ZURICH_LIFE_02

ZURICH_LIFE
ZURICH_LIFE

id	document_name \
CREDEM_VITA_02_1	Input\SFCR 2024 CREDEMVITA.pdf
CREDEM_VITA_02_2	Input\SFCR 2024 CREDEMVITA.pdf
AXA_VITA_02_01	Input\2024.12 QRT SFCR AXA MPS Assicurazioni V...
CREDAG_VITA_02_01	Input\ca_vita_sfcr_2024.pdf
CREDAG_VITA_02_02	Input\ca_vita_sfcr_2024.pdf
REALE_02_01	Input\SFCR_A123S_20241231.pdf
REALE_02_02	Input\SFCR_A123S_20241231.pdf
CARDIF_02_01	Input\SFCR_A421S_20241231.pdf
MEDIO_02_01	Input\Mediolanum_Vita_Relazione_Unica_2024.pdf
MEDIO_02_02	Input\Mediolanum_Vita_Relazione_Unica_2024.pdf
GEN_ITA_02_01	Input\2024 Relazione sulla Solvibilita e Condi...
GEN_ITA_02_02	Input\2024 Relazione sulla Solvibilita e Condi...
BMP_VITA_02_01	Input\Fascicolo-SFCR_2024_BBPMV-signed_con-opi...
BMP_VITA_02_02	Input\Fascicolo-SFCR_2024_BBPMV-signed_con-opi...
HDI_01	Input\HDI Assicurazioni S.p.A. - Avviso pubbli...
HDI_02	Input\HDI Assicurazioni S.p.A. - Avviso pubbli...
POSTE_VITA_01	Input\relazioneunicasolvibilita_condizionedefina...
POSTE_VITA_02	Input\relazioneunicasolvibilita_condizionedefina...
INTESA_VITA_01	Input\SFCR Gruppo ISPA 2024_ENG.pdf
INTESA_VITA_02	Input\SFCR Gruppo ISPA 2024_ENG.pdf
CNP_VITA_01	Input\SFCR_2024_CNP_Vita_Assicura_con_Relazion...
CNP_VITA_02	Input\SFCR_2024_CNP_Vita_Assicura_con_Relazion...
ITAS_VITA_01	Input\SFCR_G0010_20241231.pdf
ITAS_VITA_02	Input\SFCR_G0010_20241231.pdf
HELVETIA_VITA_01	Input\SFCR_Annex_Helvetia Vita_2024.pdf
HELVETIA_VITA_02	Input\SFCR_Annex_Helvetia Vita_2024.pdf
VITTORIA_01	Input\SFCR-2024_Vittoria.pdf
VITTORIA_02	Input\SFCR-2024_Vittoria.pdf
GROUPAMA_01	Input\SFCR-2024-Relazione-solvibilita-condizio...
ALLIANZ_UNICREDIT_01	Input\UAV_Fascicolo_SFCR_2024_con_relazioni_Pw...
ALLIANZ_UNICREDIT_02	Input\UAV_Fascicolo_SFCR_2024_con_relazioni_Pw...
ZURICH_LIFE_01	Input\zil_scfr_2024_web.pdf
ZURICH_LIFE_02	Input\zil_scfr_2024_web.pdf

id	table_name	page_number \
CREDEM_VITA_02_1	S.02.01.02_A	197
CREDEM_VITA_02_2	S.02.01.02_L	198
AXA_VITA_02_01	S.02.01.02_A	1
CREDAG_VITA_02_01	S.02.01.02_A	61
CREDAG_VITA_02_02	S.02.01.02_L	62
REALE_02_01	S.02.01.02_A	158
REALE_02_02	S.02.01.02_L	159

CARDIF_02_01	S.02.01.02_A	106
MEDIO_02_01	S.02.01.02_A	164
MEDIO_02_02	S.02.01.02_L	165
GEN_ITA_02_01	S.02.01.02_A	144
GEN_ITA_02_02	S.02.01.02_L	145
BMP_VITA_02_01	S.02.01.02_A	80
BMP_VITA_02_02	S.02.01.02_L	81
HDI_01	S.02.01.02_A	104
HDI_02	S.02.01.02_L	105
POSTE_VITA_01	S.02.01.02_A	136
POSTE_VITA_02	S.02.01.02_L	137
INTESA_VITA_01	S.02.01.02_A	250
INTESA_VITA_02	S.02.01.02_L	251
CNP_VITA_01	S.02.01.02_A	116
CNP_VITA_02	S.02.01.02_L	117
ITAS_VITA_01	S.02.01.02_A	233
ITAS_VITA_02	S.02.01.02_L	234
HELVETIA_VITA_01	S.02.01.02_A	2
HELVETIA_VITA_02	S.02.01.02_L	3
VITTORIA_01	S.02.01.02_A	93
VITTORIA_02	S.02.01.02_L	94
GROUPAMA_01	S.02.01.02_A	71
ALLIANZ_UNICREDIT_01	S.02.01.02_A	110
ALLIANZ_UNICREDIT_02	S.02.01.02_L	111
ZURICH_LIFE_01	S.02.01.02_A	58
ZURICH_LIFE_02	S.02.01.02_L	59

output_pdf_path \

id	
CREDEM_VITA_02_1	Single_pdf/CREDEM_S02_01_02_1_2024.pdf
CREDEM_VITA_02_2	Single_pdf/CREDEM_S02_01_02_2_2024.pdf
AXA_VITA_02_01	Single_pdf/AXA_S02_01_02_1_2024.pdf
CREDAG_VITA_02_01	Single_pdf/CA_S02_01_02_1_2024.pdf
CREDAG_VITA_02_02	Single_pdf/CA_S02_01_02_2_2024.pdf
REALE_02_01	Single_pdf/RM_S02_01_02_1_2024.pdf
REALE_02_02	Single_pdf/RM_S02_01_02_2_2024.pdf
CARDIF_02_01	Single_pdf/CARDIF_S02_01_02_1_2024.pdf
MEDIO_02_01	Single_pdf/MEDIO_S02_01_02_1_2024.pdf
MEDIO_02_02	Single_pdf/MEDIO_S02_01_02_2_2024.pdf
GEN_ITA_02_01	Single_pdf/GEN_ITA_S02_01_02_1_2024.pdf
GEN_ITA_02_02	Single_pdf/GEN_ITA_S02_01_02_2_2024.pdf
BMP_VITA_02_01	Single_pdf/BMP_VITA_S02_01_02_1_2024.pdf
BMP_VITA_02_02	Single_pdf/BMP_VITA_S02_01_02_2_2024.pdf
HDI_01	Single_pdf/HDI_S02_01_02_1_2024.pdf
HDI_02	Single_pdf/HDI_S02_01_02_2_2024.pdf
POSTE_VITA_01	Single_pdf/POSTE_VITA_S02_01_02_1_2024.pdf
POSTE_VITA_02	Single_pdf/POSTE_VITA_S02_01_02_2_2024.pdf
INTESA_VITA_01	Single_pdf/INTESA_VITA_S02_01_02_1_2024.pdf

INTESA_VITA_02	Single_pdf/INTESA_VITA_S02_01_02_2_2024.pdf
CNP_VITA_01	Single_pdf/CNP_VITA_S02_01_02_1_2024.pdf
CNP_VITA_02	Single_pdf/CNP_VITA_S02_01_02_2_2024.pdf
ITAS_VITA_01	Single_pdf/ITAS_VITA_S02_01_02_1_2024.pdf
ITAS_VITA_02	Single_pdf/ITAS_VITA_S02_01_02_2_2024.pdf
HELVETIA_VITA_01	Single_pdf/HELVETIA_VITA_S02_01_02_1_2024.pdf
HELVETIA_VITA_02	Single_pdf/HELVETIA_VITA_S02_01_02_2_2024.pdf
VITTORIA_01	Single_pdf/VITTORIA_S02_01_02_1_2024.pdf
VITTORIA_02	Single_pdf/VITTORIA_S02_01_02_2_2024.pdf
GROUPAMA_01	Single_pdf/GROUPAMA_S02_01_02_1_2024.pdf
ALLIANZ_UNICREDIT_01	Single_pdf/ALLIANZ_UNICREDIT_S02_01_02_1_2024.pdf
ALLIANZ_UNICREDIT_02	Single_pdf/ALLIANZ_UNICREDIT_S02_01_02_2_2024.pdf
ZURICH_LIFE_01	Single_pdf/ZURICH_LIFE_S02_01_02_1_2024.pdf
ZURICH_LIFE_02	Single_pdf/ZURICH_LIFE_S02_01_02_2_2024.pdf

	output_final_path \
id	
CREDEM_VITA_02_1	Output/CREDEM_S02_01_02_1_2024.csv
CREDEM_VITA_02_2	Output/CREDEM_S02_01_02_2_2024.csv
AXA_VITA_02_01	Output/AXA_S02_01_02_1_2024.csv
CREDAG_VITA_02_01	Output/CA_S02_01_02_1_2024.csv
CREDAG_VITA_02_02	Output/CA_S02_01_02_2_2024.csv
REALE_02_01	Output/RM_S02_01_02_1_2024.csv
REALE_02_02	Output/RM_S02_01_02_2_2024.csv
CARDIF_02_01	Output/CARDIF_S02_01_02_1_2024.csv
MEDIO_02_01	Output/MEDIO_S02_01_02_1_2024.csv
MEDIO_02_02	Output/MEDIO_S02_01_02_2_2024.csv
GEN_ITA_02_01	Output/GEN_ITA_S02_01_02_1_2024.csv
GEN_ITA_02_02	Output/GEN_ITA_S02_01_02_2_2024.csv
BMP_VITA_02_01	Output/BMP_VITA_S02_01_02_1_2024.csv
BMP_VITA_02_02	Output/BMP_VITA_S02_01_02_2_2024.csv
HDI_01	Output/HDI_S02_01_02_1_2024.csv
HDI_02	Output/HDI_S02_01_02_2_2024.csv
POSTE_VITA_01	Output/POSTE_VITA_S02_01_02_1_2024.csv
POSTE_VITA_02	Output/POSTE_VITA_S02_01_02_2_2024.csv
INTESA_VITA_01	Output/INTESA_VITA_S02_01_02_1_2024.csv
INTESA_VITA_02	Output/INTESA_VITA_S02_01_02_2_2024.csv
CNP_VITA_01	Output/CNP_VITA_S02_01_02_1_2024.csv
CNP_VITA_02	Output/CNP_VITA_S02_01_02_2_2024.csv
ITAS_VITA_01	Output/ITAS_VITA_S02_01_02_1_2024.csv
ITAS_VITA_02	Output/ITAS_VITA_S02_01_02_2_2024.csv
HELVETIA_VITA_01	Output/HELVETIA_VITA_S02_01_02_1_2024.csv
HELVETIA_VITA_02	Output/HELVETIA_VITA_S02_01_02_2_2024.csv
VITTORIA_01	Output/VITTORIA_S02_01_02_1_2024.csv
VITTORIA_02	Output/VITTORIA_S02_01_02_2_2024.csv
GROUPAMA_01	Output/GROUPAMA_S02_01_02_1_2024.csv
ALLIANZ_UNICREDIT_01	Output/ALLIANZ_UNICREDIT_S02_01_02_1_2024.csv
ALLIANZ_UNICREDIT_02	Output/ALLIANZ_UNICREDIT_S02_01_02_2_2024.csv

ZURICH_LIFE_01	Output/ZURICH_LIFE_S02_01_02_1_2024.csv
ZURICH_LIFE_02	Output/ZURICH_LIFE_S02_01_02_2_2024.csv

id	codes_path	leto
CREDEM_VITA_02_1	NaN	2024
CREDEM_VITA_02_2	NaN	2024
AXA_VITA_02_01	NaN	2024
CREDAG_VITA_02_01	NaN	2024
CREDAG_VITA_02_02	NaN	2024
REALE_02_01	NaN	2024
REALE_02_02	NaN	2024
CARDIF_02_01	NaN	2024
MEDIO_02_01	NaN	2024
MEDIO_02_02	NaN	2024
GEN_ITA_02_01	NaN	2024
GEN_ITA_02_02	NaN	2024
BMP_VITA_02_01	NaN	2024
BMP_VITA_02_02	NaN	2024
HDI_01	NaN	2024
HDI_02	NaN	2024
POSTE_VITA_01	NaN	2024
POSTE_VITA_02	NaN	2024
INTESA_VITA_01	NaN	2024
INTESA_VITA_02	NaN	2024
CNP_VITA_01	NaN	2024
CNP_VITA_02	NaN	2024
ITAS_VITA_01	NaN	2024
ITAS_VITA_02	NaN	2024
HELVETIA_VITA_01	NaN	2024
HELVETIA_VITA_02	NaN	2024
VITTORIA_01	NaN	2024
VITTORIA_02	NaN	2024
GROUPAMA_01	NaN	2024
ALLIANZ_UNICREDIT_01	NaN	2024
ALLIANZ_UNICREDIT_02	NaN	2024
ZURICH_LIFE_01	NaN	2024
ZURICH_LIFE_02	NaN	2024

3 Code

3.1 Credemvita S.p.A.

S.01.02.01 1

```
[164]: table_1 = initial_read("CREDEM_VITA_02_1", master_list)
```

```
[165]: table_1 = remove_space(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[166]: table_2 = initial_read("CREDEM_VITA_02_2", master_list)
```

```
[167]: table_2 = remove_space(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[168]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["CREDEM_VITA"], ["NAME", "C0010"]])
```

```
[169]: credem_02 = table
```

```
[170]: del table, table_1, table_2
```

3.2 AXA MPS Assicurazioni Vita

S.02.01.02

```
[171]: table = initial_read("AXA_VITA_02_01", master_list)
```

```
[172]: table = replace_dash_with_zero(table, "C0010")
table = remove_dot_sep(table, "C0010")
table = replace_coma_with_dot(table, "C0010")
table = remove_dolar_sep(table, "C0010")
table = remove_space(table, "C0010")
table = as_float(table, "C0010")
table = table.fillna(0)
```

```
[173]: table.columns = pd.MultiIndex.from_product([["AXA"], ["C0010"]])
```

```
[174]: axa_02 = table
```

```
[175]: del table
```

3.3 CRÈDIT AGRICOLE VITA

S.02.01.02 1

```
[176]: table_1 = initial_read("CREDAG_VITA_02_01", master_list)
```

```
[177]: table_1 = remove_dot_sep(table_1, "C0010")
table_1 = replace_coma_with_dot(table_1, "C0010")
```

```

table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)

```

S.02.01.02 2

```
[178]: table_2 = initial_read("CREDAG_VITA_02_02", master_list)
```

```
[179]: table_2 = remove_dot_sep(table_2, "C0010")
table_2 = replace_coma_with_dot(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```
[180]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.
↳from_product([["CREDIT_AGRICOLE"], ["NAME", "C0010"]])

```

```
[181]: ca_02 = table
```

```
[182]: del table, table_1, table_2
```

3.4 Società Reale Mutua di Assicurazioni

S.02.01.02 1

```
[183]: table_1 = initial_read("REALE_02_01", master_list)
```

S.02.01.02 2

```
[184]: table_2 = initial_read("REALE_02_02", master_list)
```

```
[185]: table_2 = shorten_index(table_2)

table_2 = remove_dot_sep(table_2, "C0010")
table_2 = replace_coma_with_dot(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```
[186]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["REALE_MUTUA"], ["NAME", "C0010"]])

```

```
[187]: rm_02 = table
```

```
[188]: del table, table_1, table_2
```

3.5 Cardif Vita S.p.A.

S.02.01.02

```
[189]: table = initial_read("CARDIF_02_01", master_list)
```

```
[190]: table = table[~table.index.isna()]
```

```
[191]: table = replace_dash_with_zero(table, "C0010")
table = remove_dot_sep(table, "C0010")
table = replace_coma_with_dot(table, "C0010")
table = remove_dolar_sep(table, "C0010")
table = remove_space(table, "C0010")
table = as_float(table, "C0010")
table = table.fillna(0)
```

```
[192]: table.columns = pd.MultiIndex.from_product([["CARDIF"], ["NAME", "C0010"]])
```

```
[193]: cardif_02 = table
```

```
[194]: del table
```

3.6 MEDIOLANUM VITA S.p.A.

S.02.01.02 1

```
[195]: table_1 = initial_read("MEDIO_02_01", master_list)
```

```
[196]: table_1 = remove_dot_sep(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[197]: table_2 = initial_read("MEDIO_02_02", master_list)
```

```
[198]: table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[199]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["MEDIOLANUM"], ["NAME", "C0010"]])
```

```
[200]: medio_02 = table
```

```
[201]: del table, table_1, table_2
```

3.7 Generali Italia S.p.A.

S.02.01.02 1

```
[202]: table_1 = initial_read("GEN_ITA_02_01", master_list)
```

```
[203]: table_1 = table_1.loc[table_1.index.notna()]

table_1 = remove_dot_sep(table_1, "C0010")
table_1 = replace_coma_with_dot(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[204]: table_2 = initial_read("GEN_ITA_02_02", master_list)
```

```
[205]: table_2 = table_2.loc[table_2.index.notna()]

table_2 = remove_dot_sep(table_2, "C0010")
table_2 = replace_coma_with_dot(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[206]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.
    ↪from_product([["GENERALI_ITALIA"], ["NAME", "C0010"]])
```

```
[207]: gen_ita_02 = table
```

```
[208]: del table, table_1, table_2
```

3.8 Banco BPM Vita S.p.A.

S.02.01.02 1

```
[209]: table_1 = initial_read("BMP_VITA_02_01", master_list)
```

```
[210]: table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_comma_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[211]: table_2 = initial_read("BMP_VITA_02_02", master_list)
```

```
[212]: table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_comma_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[213]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["BMP_VITA"], ["NAME", "C0010"]])
```

```
[214]: bmp_vita_02 = table
```

```
[215]: del table, table_1, table_2
```

3.9 HDI ASSICURAZIONI S.p.A.

```
[216]: table_1 = initial_read("HDI_01", master_list)
```

```
[217]: table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[218]: table_2 = initial_read("HDI_02", master_list)
```

```
[219]: table_2 = table_2[~table_2.index.isna()]
table_2 = table_2[~(table_2.index == '---')]
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[220]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["HDI"], ["NAME", "C0010"]])
```

```
[221]: hdi_02 = table
```

```
[222]: del table, table_1, table_2
```

3.10 Gruppo Assicurativo Poste Vita

S.02.01.02 1

```
[223]: table_1 = initial_read("POSTE_VITA_01", master_list)
```

```
[224]: table_1 = remove_space(table_1, "C0010")
table_1 = replace_dash_with_zero(table_1, "C0010")
```



```

table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_bracket_instead_negative(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)

```

S.02.01.02 2

```
[225]: table_2 = initial_read("POSTE_VITA_02", master_list)
```

```
[226]: table_2 = table_2.loc[table_2.index!="-"]

table_2 = remove_space(table_2, "C0010")
table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_bracket_instead_negative(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```
[227]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["POSTE_VITA"], ["NAME", "C0010"]])
```

```
[228]: poste_vita_02 = table
```

```
[229]: del table, table_1, table_2
```

3.11 FIDEURAM VITA S.P.A.

S.02.01.02 1

```
[230]: table_1 = initial_read("INTESA_VITA_01", master_list)
```

```
[231]: table_1 = table_1.loc[table_1.index.notna()]

table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_comma_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)

```

S.02.01.02 2

```
[232]: table_2 = initial_read("INTESA_VITA_02", master_list)
```

```
[233]: table_2 = table_2.loc[table_2.index.notna()]

table_2 = replace_dash_with_zero(table_2, "C0010")

```

```

table_2 = remove_space(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_comma_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```

[234]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["INTESA_VITA"], ["NAME", "C0010"]])

```

```

[235]: intesa_vita_02 = table

```

```

[236]: del table, table_1, table_2

```

3.12 CNP Vita Assicura S.p.A.

S.02.01.02 1

```

[237]: table_1 = initial_read("CNP_VITA_01", master_list)

```

```

[238]: table_1 = table_1[~table_1.index.isna()]
table_1 = convert_index_to_R(table_1)

table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)

```

S.02.01.02 2

```

[239]: table_2 = initial_read("CNP_VITA_02", master_list)

```

```

[240]: table_2 = convert_index_to_R(table_2)

table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```

[241]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["CNP_VITA"], ["NAME", "C0010"]])

```

```

[242]: cnp_vita_02 = table

```

```

[243]: del table, table_1, table_2

```

3.13 ITAS VITA

S.02.01.02 1

```
[244]: table_1 = initial_read("ITAS_VITA_01", master_list)
```

```
[245]: table_1 = table_1.loc[table_1.index.notna()]

table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[246]: table_2 = initial_read("ITAS_VITA_02", master_list)
```

```
[247]: table_2 = table_2[~table_2.index.isna()]

table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[248]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["ITAS_VITA"], ["NAME", "C0010"]])
```

```
[249]: itas_vita_02 = table
```

```
[250]: del table, table_1, table_2
```

3.14 Helvetia Vita S.p.A.

S.02.01.02 1

```
[251]: table_1 = initial_read("HELVETIA_VITA_01", master_list)
```

```
[252]: table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[253]: table_2 = initial_read("HELVETIA_VITA_02", master_list)

[254]: table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

[255]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["HELVATIA_VITA"], ["NAME", "C0010"]])

[256]: helvatia_vita_02 = table

[257]: del table, table_1, table_2
```

3.15 Vittoria Assicurazioni S.p.A.

S.02.01.02 1

```
[258]: table_1 = initial_read("VITTORIA_01", master_list)

[259]: table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_bracket_instead_negative(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[260]: table_2 = initial_read("VITTORIA_02", master_list)

[261]: table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_bracket_instead_negative(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

[262]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["VITTORIA"], ["C0010"]])

[263]: vittora_vita_02 = table

[264]: del table, table_1, table_2
```

3.16 GROUPAMA ASSICURAZIONI S.P.A.

S.02.01.02 1

```
[265]: table = initial_read("GROUPAMA_01", master_list)
```

```
[266]: table = table[~table.index.isna()]  
table = table[~(table.index == '---')]
```

```
[267]: table = replace_dash_with_zero(table, "C0010")  
table = remove_space(table, "C0010")  
table = remove_dolar_sep(table, "C0010")  
table = remove_bracket_instead_negative(table, "C0010")  
table = remove_dot_sep(table, "C0010")  
table = as_float(table, "C0010")  
table = table.fillna(0)
```

```
[268]: table.columns = pd.MultiIndex.from_product([["GROUPAMA"], ["NAME", "C0010"]])
```

```
[269]: groupama_02 = table
```

```
[270]: del table
```

3.17 UniCredit Allianz Vita S.p.A.

S.02.01.02 1

```
[271]: table_1 = initial_read("ALLIANZ_UNICREDIT_01", master_list)
```

```
[272]: table_1 = table_1.loc[table_1.index.notna()]  
  
table_1 = replace_dash_with_zero(table_1, "C0010")  
table_1 = remove_space(table_1, "C0010")  
  
table_1 = remove_dolar_sep(table_1, "C0010")  
table_1 = remove_bracket_instead_negative(table_1, "C0010")  
table_1 = remove_dot_sep(table_1, "C0010")  
table_1 = as_float(table_1, "C0010")  
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[273]: table_2 = initial_read("ALLIANZ_UNICREDIT_02", master_list)
```

```
[274]: table_2 = table_2.loc[table_2.index.notna()]  
  
table_2 = replace_dash_with_zero(table_2, "C0010")  
table_2 = remove_space(table_2, "C0010")
```

```

table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_bracket_instead_negative(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```

[275]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.
↳from_product([["ALLIANZ_UNICREDIT"], ["NAME", "C0010"]])

```

```

[276]: allianz_02 = table

```

```

[277]: del table, table_1, table_2

```

3.18 Zurich Investments Life S.p.A.

```

[278]: table_1 = initial_read("ZURICH_LIFE_01", master_list)

```

```

[279]: table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)

```

S.02.01.02 2

```

[280]: table_2 = initial_read("ZURICH_LIFE_02", master_list)

```

```

[281]: table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```

[282]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["ZURICH_LIFE"], ["NAME", "C0010"]])

```

```

[283]: zurich_life_02 = table

```

```

[284]: del table, table_1, table_2

```

3.19 Dirty master table

```

[285]: master_ita_02 = pd.read_csv("Input/S02_01_02_Master_Table.csv", index_col = 0).
↳fillna(0)
master_ita_02.columns = pd.MultiIndex.from_product([["MASTER"], ["NAME"]])

```

```

[286]: credem_02 = credem_02.drop(columns=("CREDEM_VITA", "NAME"))
#axa_02 = axa_02.drop(columns=("AXA", "NAME"))
ca_02 = ca_02.drop(columns=("CREDIT_AGRICOLE", "NAME"))
rm_02 = rm_02.drop(columns=("REALE_MUTUA", "NAME"))
cardif_02 = cardif_02.drop(columns=("CARDIF", "NAME"))

```

```

medio_02 = medio_02.drop(columns=("MEDIOLANUM", "NAME"))
gen_ita_02 = gen_ita_02.drop(columns=("GENERALI_ITALIA", "NAME"))
bmp_vita_02 = bmp_vita_02.drop(columns=("BMP_VITA", "NAME"))
hdi_02 = hdi_02.drop(columns=("HDI", "NAME"))
poste_vita_02 = poste_vita_02.drop(columns=("POSTE_VITA", "NAME"))
intesa_vita_02 = intesa_vita_02.drop(columns=("INTESA_VITA", "NAME"))
cnp_vita_02 = cnp_vita_02.drop(columns=("CNP_VITA", "NAME"))
itas_vita_02 = itas_vita_02.drop(columns=("ITAS_VITA", "NAME"))
helvatia_vita_02 = helvatia_vita_02.drop(columns=("HELVATIA_VITA", "NAME"))
#vittoria_vita_02 = vittoria_vita_02.drop(columns=("VITTORIA", "NAME"))
groupama_02 = groupama_02.drop(columns=("GROUPAMA", "NAME"))
allianz_02 = allianz_02.drop(columns=("ALLIANZ_UNICREDIT", "NAME"))
zurich_life_02 = zurich_life_02.drop(columns=("ZURICH_LIFE", "NAME"))

```

```

[287]: table_02 = master_ita_02.join(credem_02).join(axa_02).join(ca_02).join(rm_02).
      ↪join(cardif_02).join(medio_02).join(gen_ita_02).join(bmp_vita_02).
      ↪join(hdi_02).join(poste_vita_02).join(intesa_vita_02).join(cnp_vita_02).
      ↪join(itas_vita_02).join(helvatia_vita_02).join(vittoria_vita_02).
      ↪join(groupama_02).join(allianz_02).join(zurich_life_02)

```

```

[288]: table_02 = table_02[~table_02.index.duplicated(keep=False)]

```

```

[289]: table_02.to_csv("Dirty_Combined/Table_ita_S02.csv")

```